Journal of Industrial Intelligence
https://www.acadlore.com/journals/JII

Prioritizing Cold Supply Chain Barriers: A q-Rung Orthopair Fuzzy Decision Framework

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Received: 05-01-2025 **Revised:** 06-15-2025 **Accepted:** 06-30-2025

Citation: S. Korucuk, A. Aytekin, and A. Güngör, "Prioritizing cold supply chain barriers: A q-Rung Orthopair Fuzzy decision framework," *J. Ind Intell.*, vol. 3, no. 3, pp. 137–145, 2025. https://doi.org/10.56578/jii030301.

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Abstract: For businesses, the effective management of cold supply chains is critical to minimizing food losses and ensuring customer satisfaction. Identifying and prioritizing the obstacles that disrupt these processes is therefore a strategic necessity. However, existing literature largely addresses cold supply chain challenges in a fragmented manner, lacking systematic prioritization frameworks that account for the inherent uncertainty and subjective judgments present in real-world operations. To address this deficiency, this study proposes a structured decision framework based on the q-Rung Orthopair Fuzzy (q-ROF) Subjective Weighting Approach. This method effectively captures uncertainty and integrates expert evaluations to determine the relative importance of key cold chain barriers. Through an empirical application involving logistics managers, the framework ranks the identified obstacles to support operational and strategic decision-making. The findings reveal that Time Constraint is the most critical obstacle, directly impacting operational efficiency and customer satisfaction. In contrast, Temperature-Controlled Vehicle Cost is identified as a lower-priority factor in strategic resource allocation. These results offer a clear prioritization scheme that enables managers to focus resources on the most impactful areas, enhancing resilience and efficiency in cold chain operations. This study contributes a robust, uncertainty-aware methodology for barrier prioritization, providing actionable insights for supply chain practitioners and establishing a foundation for future research in cold chain management.

Keywords: Cold supply chain; Barrier prioritization; q-Rung Orthopair Fuzzy sets; Multi-criteria decision making; Decision framework

1 Introduction

In recent years, ensuring access to healthy and high-quality food and guaranteeing food safety has become an increasingly important issue. Alongside international organizations, governments, local authorities and individuals have also begun to express serious concern about these issues due to developments such as rapid population growth, increasing urbanization, climate change, global warming and environmental pollution [1]. These developments have highlighted the need for sustainability and efficiency in food production, distribution and supply chain management processes, while also bringing to the fore fundamental objectives such as the effective utilization of food resources, the reduction of losses and the provision of safe food supplies. Rising demand levels and increasing environmental pressures are driving businesses to develop more careful and innovative approaches both in strategic planning and operational implementation [2].

Cold chain logistics in particular plays an extremely important economic role. It is crucial that the cold chain, one of the fundamental elements of the supply chain, continues without interruption [3]. The cold chain encompasses activities related to the storage, distribution, handling, transportation, and continuous monitoring of temperature-sensitive products. In addition, it refers to an integrated system consisting of equipment such as cold storage facilities, refrigerators, transport containers and specially equipped vehicles, as well as the necessary records, procedures and monitoring mechanisms to ensure that products are protected safely and to high quality standards. This comprehensive structure prevents quality loss by ensuring that products are stored under appropriate temperature conditions and also guarantees compliance with food safety standards [4]. The cold supply chain refers to a supply chain structure that aims to protect the quality and safety of products that must be kept within a specific temperature range throughout all

stages, from production to delivery to the end user. This system requires the continuous monitoring and control of temperature conditions to prevent deterioration in the physical, chemical and microbiological properties of products. In this context, maintaining appropriate temperature conditions during the production, storage, distribution and consumption processes of frozen foods, fresh fruit and vegetables, medicines and other perishable products is of great importance. Effective management of the cold supply chain contributes to ensuring food safety, minimising product losses and increasing customer satisfaction [5]. According to another definition, this concept refers to a process in which all operations, from the storage of raw materials to the distribution of final products, are carried out in a temperature-controlled environment. This approach aims not only to maintain product quality but also to reduce the likelihood of spoilage and ensure compliance with food safety standards [6]. In addition to supply chain operations, the personnel involved in the process and the equipment used must be planned and effectively managed to ensure temperature control at every stage [7].

With the increase in population, demand for perishable goods has also risen significantly. This growing need necessitates the development and expansion of cold chain infrastructure, such as packaging facilities, refrigerated transport systems and cold storage facilities, in order to maintain product quality and safety [8]. The proper planning and effective management of this infrastructure is of vital importance in terms of reducing food losses and improving the overall performance of the supply chain. Investments in cold chain systems are therefore of strategic value in terms of preventing food waste and improving overall supply chain efficiency [9].

However, cold supply chains also present considerable environmental challenges. It is essential to store and transport products at low temperatures close to or below freezing point, which necessitates the intensive use of refrigerated warehouses, cold storage facilities, and refrigerated transport vehicles. These systems consume high levels of energy for cooling processes, and the increasing energy demand is directly linked to higher CO_2 emissions, particularly from fossil fuel-based energy production, thereby exacerbating the adverse effects of climate change. In addition, hydrofluorocarbon gases, which are widely used in refrigeration systems, pose serious environmental threats due to their high global warming potential and long atmospheric lifetime. Routine or unexpected leaks of hydrofluorocarbon gases throughout cold supply chain processes significantly increase global warming and pose significant risks to environmental sustainability. Therefore, improving energy efficiency, evaluating alternative refrigerants, and implementing measures to prevent refrigerant leaks are fundamental issues that must be carefully addressed in the design and operation of cold supply chains [10].

In this context, the difficulties arising in cold supply chains should not be viewed merely as technical or administrative problems affecting businesses' daily operations; rather, they should be assessed as multifaceted and strategic elements that concern all segments of society. These issues are directly linked not only to ensuring food safety and protecting public health, but also to the efficient use of natural resources, strengthening environmental sustainability, and maintaining economic stability. Disruptions in cold supply chains can negatively impact businesses' competitive advantage and overall performance levels by leading to increased product losses, rising costs, and disruption to supply continuity. Furthermore, key performance metrics such as sustainability, cost control, operational efficiency, effectiveness, and customer satisfaction are also directly affected by these issues.

Given the critical importance of these challenges, systematically identifying, thoroughly evaluating, and prioritising the difficulties encountered in cold supply chains is essential for ensuring effective supply chain management and securing the long-term success of businesses. Addressing these issues is a strategic imperative for businesses aiming to transform their processes into agile, flexible and resilient structures capable of adapting to variable market conditions. Reducing disruptions in cold supply chains not only enables more efficient use of resources but also secures supply continuity, delivering benefits at both the economic and societal levels. Effectively managing these challenges contributes to reducing operational risks, enhancing service quality, and improving overall supply chain performance.

A comprehensive review of the literature reveals that studies specifically focusing on barriers in cold supply chains are quite limited. The majority of existing research is confined to specific sectors or countries and does not provide an in-depth assessment of the root causes of these obstacles, their interrelationships, or their overall impact on supply chain performance. This highlights a significant gap in the literature and points to a need for applied research in this area.

To address this deficiency, this study aims to systematically identify, evaluate, and prioritize the obstacles encountered in cold supply chains. The research employs the q-Rung Orthopair Fuzzy (q-ROF) Subjective Weighting Approach, a multi-criteria decision-making (MCDM) technique that allows for a more accurate and systematic determination of importance levels by taking into account uncertainty and subjective assessments. Within the scope of the research, empirical data were collected from logistics managers operating in a region with significant cold chain activity, and the identified barriers were examined and systematically categorized according to their relative importance. This methodological framework enables the execution of a systematic and transparent review process by simultaneously considering multiple criteria and integrating expert assessments into the analysis.

In addition to empirical findings, the study establishes a solid conceptual and methodological foundation by

comprehensively evaluating the theoretical background, fundamental concepts, and existing literature related to cold supply chain applications. The remainder of the paper is organized as follows: Section 2 presents a comprehensive literature review; Section 3 describes the methodology; Section 4 reports the results; and Section 5 concludes with a discussion of findings, managerial implications, and directions for future research.

2 Literature

The existing literature offers a variety of perspectives on cold supply chain management. This section reviews relevant national and international studies that contribute to strengthening the theoretical basis of the current research by presenting important findings on cold supply chain applications, encountered difficulties, performance measurement approaches, and sustainability dimensions.

Early research in the field focused primarily on the technical and operational aspects of cold chain systems. James et al. [11] comprehensively addressed the theoretical and practical elements that producers and suppliers should consider when determining refrigeration systems for the food supply chain. Their study analyzed criteria such as the efficiency of refrigeration technologies, energy consumption, cost components, and product quality preservation, offering important insights for selecting the most suitable refrigeration system. Around the same period, Faisal [12] conducted an extensive literature review on cold supply chains and developed a hierarchical model for supply chain agility. The study systematically analyzed the main dimensions of agile supply chains and the interactions between these dimensions, revealing that enhancing agility is vital for managing uncertainties and improving responsiveness.

As research progressed, scholars began to pay greater attention to the environmental dimensions of cold chain operations. Putri et al. [13] performed a comparative analysis of the environmental impacts of traditional supply chain and cold chain management practices, putting forward various recommendations for improving distribution activities. Their study highlighted the differences between the two systems, particularly in terms of energy consumption, emission rates, and environmental sustainability dimensions. Laguerre et al. [14] took a methodological approach, analyzing deterministic modelling techniques for cold chain equipment and examining their applicability in food processing and cold chain operations. The study recommended the combined use of deterministic and stochastic modelling to manage uncertainties more effectively and improve operational decision-making.

Another stream of research has focused on traceability, monitoring, and technological innovation. Asadi and Hosseini [15] examined traceability and tracking tools used to ensure the quality and safety of basic agricultural products throughout the cold supply chain. Their research emphasized the importance of information technologies and real-time monitoring systems in maintaining temperature control and ensuring transparency across various stages, highlighting the contribution of effective traceability mechanisms to reducing risks and strengthening regulatory compliance. Shashi and Singh [16] took a broader perspective, comprehensively reviewing previous studies on all stages of agricultural product supply chains. Their work addressed the challenges encountered at each stage, issues related to efficiency, and sustainability factors, offering recommendations for improving overall supply chain performance.

Technological advancements have also driven innovation in cold chain monitoring. Xiao et al. [17] developed a Temperature Monitoring System based on a Wireless Sensor Network (WSN) integrated with data compression and transmission features for frozen and chilled seafood products. This system provides real-time temperature monitoring throughout the cold supply chain, while the data compression feature enhances energy efficiency and minimizes data loss. In the context of public health, Ashok et al. [18] analyzed initiatives implemented in ten different countries aimed at improving vaccine cold chains. Their research identified the root causes of three common problems limiting cold chain performance and offered potential solutions, highlighting the importance of systematically addressing barriers to cold chain management.

MCDM approaches have also been applied to cold chain logistics. Korucuk et al. [19] applied the Analytic Hierarchy Process and Grey Relational Analysis techniques to select third-party logistics (3PL) companies for cold chain transportation firms operating in Istanbul. The performance of 3PL providers was evaluated using an MCDM approach, providing a more systematic and objective supplier selection process. Similarly, Aygün and Çağil [20] examined cold chain applications in the food sector by solving two different vehicle routing problems under specific vehicle and cost constraints, evaluating the efficiency and cost optimization of cold chain transportation.

Environmental sustainability has emerged as a central theme in more recent studies. Babagolzadeh et al. [21] examined the effects of carbon emissions associated with storage and transport in cold supply chains under carbon tax applications and uncertain demand conditions, emphasizing the importance of sustainable logistics practices and noting that emission reduction strategies could contribute to both lowering environmental impacts and increasing cost efficiency. İpekçi and Tanyas [22] took a comparative perspective, examining current cold chain practices worldwide and outlining both the status of these practices and the overall outlook for the sector in Turkey, contributing to identifying the sector's strengths and areas for improvement.

Risk assessment has also received considerable attention. Khan et al. [23] conducted a comprehensive literature

review and identified 40 different risks associated with cold chains in the context of a developing country. The study systematically addressed the types, sources, and potential impacts of these risks throughout the supply chain, providing valuable insights for developing preventive strategies. Meanwhile, Turgut [24] conducted a comprehensive review of new generation technologies in cold chain logistics, addressing the integration of advanced technologies such as the Internet of Things (IoT), artificial intelligence, sensor technologies, and data analytics into cold chain processes, and evaluating their impact on operational efficiency, traceability, and supply chain transparency.

More recently, scholars have continued to assess the evolving landscape of cold chain operations. Bentyn and Konecka [25] comprehensively assessed the adequacy and effectiveness of existing cold chain technologies, taking into account rising temperatures, increasing demand, and necessary environmental protection measures. Fatorachian and Pawar [26] analyzed the key determinants of demand trends in medium-scale cold chain operations, examining the effects of demand forecasts on supply chain planning and resource allocation, and exploring which factors should be prioritized to increase operational efficiency for medium-sized enterprises.

A comprehensive review of the literature reveals that studies specifically focusing on barriers in cold supply chains remain quite limited. The majority of existing research is confined to specific sectors or countries and does not provide an in-depth assessment of the root causes of these obstacles, their interrelationships, or their overall impact on supply chain performance. This highlights a significant gap in the literature and points to a clear need for applied research in this area. Therefore, a study that systematically classifies and prioritizes these barriers while also proposing effective solution strategies could provide valuable contributions to both academic knowledge and sectoral applications. Such research would help businesses make their cold chain operations more efficient, flexible, and sustainable, while also offering a solid foundation for future work in the field.

3 Methodology

This study employs a q-ROF subjective weighting approach. In this context, the prerequisites for q-ROFs are given first. Assume that X be a nonempty set. A q-ROFs A of X is given in Eq. (1).

$$A = \{\langle x, (aA(\mathrm{x}), bA(\mathrm{x}))\rangle \mid \mathrm{x} \in \mathrm{X}\} \tag{1}$$

where, $aA(\mathrm{x})$ is the membership degree and $bA(\mathrm{x})$ is the non-membership degree.

The conditions $aA(\mathrm{x}) \in [0, 1]$, $bA(x) \in [0, 1]$, $0 \leq (aA(x)q + bA(\mathrm{x})q) \leq 1$, $(q \geq 1)$ should be satisfied in q-ROFs. The uncertainty membership degree is defined as $\pi A(\mathrm{x}) = (aA(x)q + bA(\mathrm{x})q - aA(x)qbA(\mathrm{x})q)1/q$. For convenience, let $g1 = \langle a1, b1\rangle$ and $g2 = \langle a2, b2\rangle$ be two q-ROF numbers (q-ROFNs). The fundamental operations, the score function $(\mathcal{S}(g1))$, and the accuracy function $(\mathcal{A}(g1))$ are presented in Eqs. (2)–(8) and studies [27–29].

$$g_1 \oplus g_1 = \left(\sqrt[q]{(a_1)^q + (a_2)^q - (a_1)^q (a_2)^q}, b_1 b_2\right) \tag{2}$$

$$g_1 \otimes g_2 = \left(a_1 a_2, \sqrt[q]{(b_1)^q + (b_2)^q - (b_1)^q (b_2)^q}\right) \tag{3}$$

$$\xi g_1 = \left(\sqrt[q]{1 - (1 - (a_1)^q)^{\xi}}, (b_1)^{\xi}\right), \xi \geq 0 \tag{4}$$

$$g_1^{\xi} = \left((a_1)^{\xi}, \sqrt[q]{1 - (1 - (b_1)^q)^{\xi}}\right), \xi \geq 0 \tag{5}$$

$$(g_1)^c = (b_1, a_1) \tag{6}$$

$$\mathcal{S}(g_1) = \frac{1}{2}\left((a_1)^q - (b_1)^q + 1\right), \quad S(g_1) \in [0, 1] \tag{7}$$

$$\mathcal{A}(g_1) = (a_1)^q + (b_1)^q, \quad \mathcal{A}(g_1) \in [0, 1] \tag{8}$$

Assume that $g_j = \langle a_j, b_j\rangle$ is a collection of q-ROFNs. j = 1, …, n. The q-ROF weighted arithmetic average (q-ROFWAA) operator and the q-ROF weighted geometric average (q-ROFWGA) operator provided in Eqs. (9) and (10), respectively in studies [27–29].

$$\begin{aligned} &\mathrm{q-ROFWAA}(g_j) \\ &= \left\langle \left(-\prod_{\mathrm{j}=1}^{n} \left(-a_j^q\right)^{w_j}\right), \prod_{\mathrm{j}=1}^{n} b_j^{w_j} \right\rangle \end{aligned} \tag{9}$$

$$\begin{aligned} &\mathrm{q-ROFWGA}\left(g_j\right) \\ &= \left\langle \prod_{\mathrm{j}=1}^{n} a_i^{w_j},\ \left(-\prod_{\mathrm{j}=1}^{n} \left(-b_j^q\right)^{w_j} \right) \right\rangle \end{aligned} \tag{10}$$

Here $W = (w_1, \ldots, w_n)^T$ is the weight vector, $0 \leq w_j \leq 1$, and $\sum_{j=\mathrm{n}}^{n} w_i = 1$. Following the clarification of the basic operations, the next part will outline the q-ROF-Subjective Weighting Approach [30].

Step 1. The lists of criteria and experts are identified. In this context, $C = \{C_1, \ldots, C_j, \ldots C_n\}$ represents the list of criteria, while $E = \{E_1, \ldots, E_k, \ldots E_z\}$ represents the list of experts.

Step 2. The weight coefficient of each expert λ_k is determined. $k = 1, \ldots, z$, $0 \leq \lambda_k \leq 1$ and $\sum_{k=1}^{z} \lambda_k = 1$.

Step 3. The importance of criteria is evaluated by experts using linguistic terms given in Table 1.

Table 1. The linguistic terms and their corresponding q-Rung Orthopair Fuzzies (q-ROFs) equivalents

Linguistic Terms	Notations	q-ROFNs	
		a	b
Absolutely high	AH	0.99	0.20
Very very high	VVH	0.91	0.28
Very high	VH	0.83	0.36
High	H	0.75	0.44
Medium high	MH	0.67	0.52
Medium	M	0.60	0.60
Medium low	ML	0.52	0.67
Low	L	0.44	0.75
Very low	VL	0.36	0.83
Very very low	VVL	0.28	0.91
Absolutely low	AL	0.20	0.99

The importance ratings of k-th expert for j-th criterion is denoted as $\iota_j^{(k)} = \left(a_j^{(k)}, b_j^{(k)}\right)$.

Step 4. The q-ROF integrated importance of j-h criterion ι_j is obtained via Eq. (9).

Step 5. The score function given in Eq. (7) is applied to determine the crisp values of ι_j.

Step 6. The subjective weight coefficient of each criterion is calculated by applying Eq. (11).

$$w_j = \frac{\delta\left(\iota_j\right)}{\sum_{j=1}^{n} \delta\left(\iota_j\right)} \text{ where } 0 \leq w_j \leq 1 \text{ and } \sum_{j=1}^{n} w_j = 1 \tag{11}$$

4 Results

Six operations managers evaluated the importance of the criteria identified in the study. These evaluations were conducted to determine the relative significance and impact of each criterion on cold supply chain operations. The collected data serve as a fundamental basis for calculating the relative weights of the criteria and for conducting multi-criteria analyses in a reliable manner. Furthermore, these assessments, based on the managers' expertise and practical experience, enhance the validity of the analysis in real-world applications and strengthen the applicability of the results for the industry. The detailed evaluations are presented in Table 2.

Table 2. The list of criteria

Criteria	Source	Notations
Energy costs	[1]	C1
Lack of infrastructure	[26]	C2
Inadequate facilities	[31]	C3
Lack of effective distribution planning	[32]	C4
Temperature controlled vehicle cost	[27]	C5
Time constraint	[16]	C6
Humidity and pH	[33, 34]	C7
Other factors (packaging conditions, materials, and storage conditions)	[1, 19]	C8

Table 3. The linguistic importance assessments of criteria provided by experts

	C1	C2	C3	C4	C5	C6	C7	C8
E1	H	H	M	M	M	M	M	M
E2	H	M	H	L	L	H	M	H
E3	H	H	M	H	M	H	H	M
E4	H	H	M	L	L	H	H	M
E5	VL	L	L	M	M	H	H	M
E6	H	H	H	M	M	VH	H	M

Note: VH = Very high, H = High, M = Medium, L = Low, VL = Very low, C1 = Energy costs, C2 = Lack of infrastructure, C3 = Inadequate facilities, C4 = Lack of effective distribution planning, C5 = Temperature controlled vehicle cost, C6 = Time constraint, C7 = Humidity and pH, C8 = Other factors (packaging conditions, materials, and storage conditions).

In this study, six experts with high knowledge and experience in the subject were consulted to solve the problem. The assessments provided by the experts are displayed in Table 3.

The calculations regarding q-ROF-Subjective Weighting Approach was carried out, and the results given in Table 4 were obtained.

Table 4. The weighting results

	C1		C2		C3		C4	
	a	b	a	b	a	b	a	b
ι_j	0.718	0.489	0.700	0.506	0.648	0.561	0.599	0.613
$\mathcal{S}(\iota_j)$	0.627		0.606		0.548		0.492	
Rank	2		4		5		7	
w_j	0.137		0.133		0.120		0.108	
	C5		**C6**		**C7**		**C8**	
	a	b	a	b	a	b	a	b
ι_j	0.558	0.646	0.749	0.448	0.711	0.487	0.634	0.569
$\mathcal{S}(\iota_j)$	0.452		0.665		0.622		0.534	
Rank	8		1		3		6	
w_j	0.0994		0.1463		0.1367		0.1176	

Note: C1 = Energy costs, C2 = Lack of infrastructure, C3 = Inadequate facilities, C4 = Lack of effective distribution planning, C5 = Temperature controlled vehicle cost, C6 = Time constraint, C7 = Humidity and pH, C8 = Other factors (packaging conditions, materials, and storage conditions).

The results presented in Table 4 reveal that the most critical criterion is C6, Time Constraint. This finding demonstrates that time plays a vital role in cold supply chain operations and directly impacts efficiency, effectiveness, and delivery schedules. Taking time constraints into account is of great importance in ensuring that orders are fulfilled on time, optimizing inventory management, and reducing delays throughout the supply chain.

Furthermore, the relationship between time constraints and other criteria, as well as their overall impact on supply chain performance, emphasizes that this factor should be prioritized in strategic planning and operational decision-making processes. Accordingly, time constraints should be considered a fundamental focus area for risk management and process improvement strategies in cold supply chain management.

5 Conclusions

Challenges in the supply chain can reduce business competitiveness and flexibility while also limiting customer satisfaction. In cold supply chains, these challenges are particularly critical, as they increase operational costs and negatively affect overall efficiency. Maintaining products within specific temperature ranges is essential, and any disruption can directly impact product quality and consumer trust. Furthermore, inefficiencies in cold supply chains extend beyond individual businesses, potentially causing inventory disruptions, transportation delays, product losses, and financial damages. Such disruptions can hinder the supply of essential temperature-sensitive goods, including food and pharmaceuticals, leading to significant economic and social consequences. Therefore, systematically identifying, analyzing, and prioritizing the challenges in cold supply chains is of vital importance for improving business performance and minimizing broader economic and societal impacts.

In this context, this study aimed to systematically identify, analyze, and prioritize the existing barriers in cold supply chain businesses with a corporate identity operating in the region studied. The research findings indicate that

the most critical challenges in the cold supply chain, in order of importance, are "Time Constraint," "Energy Costs," and "Humidity and pH." These factors are fundamental for both maintaining product quality and ensuring the timely and efficient execution of operations.

Time constraints emerge as a key issue for ensuring on-time order fulfillment, optimizing delivery schedules, managing inventory effectively, and maintaining customer satisfaction. Energy costs have a direct impact on the sustainability, cost-effectiveness, and environmental footprint of cold chain operations. Humidity and pH control, in particular, play a critical role in preserving the quality and safety of temperature- and environment-sensitive products, such as food and pharmaceuticals. The study also found that the least significant barriers are "Temperature-Controlled Vehicle Costs" and "Lack of Effective Distribution Planning." These findings provide businesses with valuable insights for prioritizing cold supply chain management activities, allocating resources more efficiently, optimizing operational decision-making processes, and controlling costs. Moreover, the results serve as a strategic guide not only at the business level but also across the entire supply chain for managing risks and improving processes. The findings provide a solid foundation for future research and the development of applied strategies, offering significant contributions to both academic literature and industry practice.

Although these factors can be partially managed within current systems, they can still indirectly affect overall supply chain performance. Therefore, effectively addressing these challenges in the cold supply chain is essential for businesses to regain competitiveness, enhance operational flexibility, and improve customer satisfaction. The adoption of energy-efficient technologies, modernization of transportation vehicles, integration of digital solutions to improve process traceability and transparency, and providing technical training to personnel are among the effective strategies to mitigate these barriers.

In addition, continuously monitoring and improving operational processes can enable the early identification of risks and the implementation of preventive measures. The application of technological investments and training programs throughout the cold chain not only helps control costs but also contributes to maintaining product quality and ensuring safety. In this way, businesses can achieve sustainability objectives while minimizing the economic and social impacts caused by supply chain disruptions.

For future research, the impact of these barriers on business performance in cold supply chains can be analyzed in greater detail using quantitative research methods. Such analyses can reveal both the direct and indirect effects of these obstacles on operational efficiency, cost control, product quality, and customer satisfaction. In addition, future research could develop strategic recommendations to ensure the sustainability of cold supply chain management. These recommendations may include more efficient use of resources, increased energy efficiency, strengthened risk management processes, and improved technological infrastructure. Thus, while providing valuable contributions to the academic literature, practical guidance can be offered to businesses on improving operational efficiency, reducing costs, and implementing sustainable practices.

This study systematically identifies the challenges encountered in cold supply chains, fills an important gap in the literature, and serves as a guiding reference for both academics and practitioners. The findings obtained from businesses operating in the region studied provide unique and valuable insights into cold chain management at a regional level, offering guidance for improving operational processes and making strategic decisions. Prioritizing critical factors such as time constraints, energy costs, and environmental conditions contributes not only to maintaining product quality but also to enhancing operational efficiency.

Future research could examine this topic using various MCDM methods, allowing for a more comprehensive analysis of cold supply chain barriers. In addition, applying these methods within the framework of different fuzzy set approaches and comparing the results obtained will enable a more reliable, objective, and comprehensive assessment of the effects of these challenges. Future research should also expand its geographical scope, compare different sectors, and investigate the impact of the IoT, big data, artificial intelligence, and other innovative digital technologies.

Such efforts could facilitate the development of a sustainable, efficiency-oriented, and technologically integrated cold supply chain structure. Consequently, these studies would provide valuable contributions to academic literature while simultaneously enhancing operational efficiency, flexibility, and sustainability for businesses.

Author Contributions

Conceptualization, S.K. and A.A.; methodology, A.A.; software, A.A.; validation, S.K., A.A., and A.G.; formal analysis, S.K.; investigation, A.Y.; resources, A.G.; data curation, S.K. and A.G.; writing—original draft preparation, S.K.; writing—review and editing, S.K. and A.A.; visualization, A.A.; supervision, S.K.; project administration, S.K. and A.A.; funding acquisition, A.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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